

Contents

Synthetic DCE — Methods & Results Brief	1
1. The one-paragraph version	1
2. Data	1
3. Pipeline	2
4. Metrics — what each one actually means	2
5. Headline results	3
6. The main scientific finding	3
7. Credibility: what was wrong and is now fixed	4
8. What we can and cannot claim	4
9. Questions you should expect	5
10. What's next	5

Synthetic DCE — Methods & Results Brief

Presentation companion. METHODS.md is the full technical record; this is what you need in the room, plus the numbers you'll be asked to defend.

Status 2026-08-06. Cohort and pipeline are settled and audited. Single-phase (Tier-1) results are in and trustworthy. The multi-timepoint (Tier-2) track is scoped and staged for, not yet run.

1. The one-paragraph version

We predict a post-contrast DCE volume from three non-contrast sequences (T2w, DWI, ADC) on 3,667 UCSF prostate exams, to avoid gadolinium in patients who can't receive it and to harmonize DCE across protocols. A 3D conditional GAN reaches **roi_pearson 0.48** inside the gland, against **-0.04** for the strongest trivial baseline (copying the T2w channel), so the model is genuinely predicting enhancement structure rather than passing anatomy through. The substantive finding is a decomposition: **within-gland structure is predictable, per-patient enhancement amplitude is not** — and amplitude is 51% of the signal. That is a property of the task, not a tuning failure, and it scopes what synthetic DCE can be used for.

2. Data

Source. Single-center UCSF, two secured mounts: anatomy/masks on DS2 (5,962 patients), 4D DCE + per-phase timing on DS3. All volumes pre-registered into T2w space.

Cohort construction — quote these numbers, they will be asked for:

stage	criterion	remaining
patients with DCE	—	5,439
successfully staged	inputs resolvable, mask matches	4,975
	DCE grid	
enhancement QC at staging	max enhancement over series >= 1.5x	4,246
DWI b-value filter	DWI resolves to b >= 600	3,684
enhancement QC recorded	—	3,667

Split **patient-level**: 2,805 train / 312 val / 550 test.

Two cohort properties worth reporting as findings, not footnotes:

- **~12–13% of patients never enhance at any timepoint.** Not a processing failure — a genuine property (failed or mistimed injection). Belongs in the data section of any paper.
- **13% of DWI resolves to b50**, which carries almost no diffusion weighting and reads T2-like. That's a *different contrast* in the same input channel as the b1000 the majority get. Excluding it costs 13% of the data and is a clean one-flag ablation. Also: 43 cases use *computed* (`_synth_`) high-b images rather than acquired ones.

Target phase chosen by elapsed time, not index. Protocols are heterogeneous — 26–70 phases at 1.7–13 s cadence — so a fixed index is a different physiological moment per patient. Measured enhancement curve: flat to ~45 s, wash-in 55–90 s (1.12→2.00×), then a **plateau of 2.1–2.5× from ~90 s past 300 s**. There is no sharp peak, so intensity-argmax lands on plateau noise at ~280 s. We take **t = 120 s**: inside the plateau, past the timing-sensitive wash-in, ~92% of maximum. Realized times cluster at 104–125 s.

Two protocol complications handled explicitly: **interleaved series** (Dixon water/fat stacked per timepoint — 27 cases; a plain index alternates between sub-series and can land on the non-enhancing one) and **corrupt timestamps** (155 cases re-selected from the enhancement curve at plateau onset).

3. Pipeline

bpMRI (T2w, DWI, ADC) → resample to common physical grid → normalize to [-1,1]
→ crop 90 × 90 × 96 mm → model → DCE prediction

Geometry — the fix that mattered most. `--spatial-size` originally meant a *pixel count*. UCSF in-plane resolution spans 0.176–0.781 mm across 256/512/672/704/736/1024 matrices, so a fixed 256-px crop covered anywhere from **40 mm to 320 mm** of anatomy — an 8× range of scale, with 172 cases cropped to 45 mm, *smaller than a prostate*. Now `--reference iso` resamples everyone to a common grid first, so the crop is a fixed **physical** field of view (90 × 90 × 96 mm) for every patient.

Normalization is per-image (single center, so no cross-scanner alignment to do): DCE robust-scaled to the body-tissue median ± 1 spread, T2w percentile, ADC fixed physical clip, DWI foreground z-score. No Nyul.

Loss. L1 + (1–SSIM) + perceptual (frozen 3D MedicalNet), with **10× ROI weighting** — the prostate is ~5% of the crop, so an unweighted loss is dominated by background.

4. Metrics — what each one actually means

Audiences conflate these. Worth a slide.

metric	measures	target
roi_pearson	within-gland co-localization — is the enhancement in the right <i>place</i>	higher
ssim_roi / mae_roi / psnr_roi	fidelity inside the gland	higher / lower / higher
roi_var_ratio	heterogeneity preserved (<1 = over-smoothed, >1 = over-textured)	1.0
roi_grad_ratio	fine detail preserved	1.0
roi_w1	ROI intensity-histogram distance	0
FID	distributional realism over 2D slices	lower

metric	measures	target
p75_corr	cross-patient enhancement tracking	higher

Two framing points to make explicitly:

- **Global SSIM/PSNR are ~99% background** and near-meaningless here. The gap between global and ROI metrics is the real signal.
- **Faithfulness and realism are different axes and can move in opposite directions.** A model can be more “realistic” and less correct. Report both.

5. Headline results

3D models, **best checkpoint**, held-out test (n = 550):

	3D GAN	3D flow (MedVAE)
roi_pearson	0.482	0.369
ssim_roi	0.397	0.276
mae_roi	0.236	0.269
roi_var_ratio (->1)	0.912	1.711
roi_grad_ratio (->1)	0.711	1.059
realism_score	0.845	0.714
PZ / TZ roi_pearson	0.339 / 0.522	0.169 / 0.417

VAL ~ TEST throughout — no overfitting.

The GAN wins on faithfulness across four independent setups (2D and 3D, pixel and latent). The flow’s one advantage is texture (grad_ratio 1.06 vs 0.71) — it is *less smooth*, which reads as more realistic while localizing worse. That is the realism/faithfulness tradeoff in one row.

Peripheral zone is roughly half as good as transition zone (0.339 vs 0.522). Report this — PZ is where most clinically significant cancer arises and where DCE drives the PI-RADS upgrade rule, so it is the clinically weakest spot. It is also the honest limitation a reviewer will find if you don’t.

6. The main scientific finding

Baseline audit, n = 550 test, 3D GAN:

predictor	MAE_roi	roi_pearson	what it knows
const (cohort mean)	0.2510	—	nothing
t2w copy	0.3372	−0.043	identity baseline
MODEL	0.2355	+0.482	trained generator
level (oracle brightness)	0.1851	—	correct per-case brightness, no structure

Read it in two halves.

Structure is learned. Copying T2w scores ~ 0 , so 0.482 is real localization — the model is not laundering the anatomy it was given.

Amplitude is not. The model tracks per-patient brightness at only $r = 0.334$ and compresses between-case spread **2.4×** (target sd 0.231 $\rightarrow$ predicted 0.098). Since **51% of ROI variance is between-case amplitude**, both models lose to a baseline that knows only the correct brightness — and the flow is worse than a *constant* predictor on MAE while still having genuine localization.

Why this is a property of the task. Per-case amplitude depends on gadolinium dose, injection rate, cardiac output, renal clearance and scanner gain. None of that is visible in T2w/DWI/ADC. Within-gland structure is. We checked the DICOM headers for the protocol variables — **they were removed by de-identification** (dose, rate, patient weight and field strength absent; manufacturer overwritten with "NA"). So the information is not merely unused, it is not present.

And it may not matter clinically. PI-RADS curve types (I progressive / II plateau / III washout) are **scale-invariant** — multiplying a curve by any constant leaves the type unchanged. The component we can't predict is orthogonal to the endpoint that matters. That converts a negative result into a scoping argument.

7. Credibility: what was wrong and is now fixed

Be upfront about this; it is a strength, not a liability.

fix	evidence
Checkerboard artifact in all three decoders (transposed convs)	lag-1 autocorrelation of the first difference -0.60 in predictions (white noise is -0.50) vs +0.81 for real DCE; FID 175 $\rightarrow$ 89 after the fix
Physical FOV instead of pixel count	8× scale range $\rightarrow$ uniform 90 mm
Best-checkpoint evaluation	3D metrics previously scored the <i>final</i> epoch; roi_pearson 0.419 $\rightarrow$ 0.482
Batch-pooled ROI metrics	pooling reported roi_pearson +0.977 on pure noise where truth is +0.020
Harmonizer leakage	Nyul was fit on the full cohort incl. test (never triggered by our configs, now impossible)
Wrong targets in training	155 cases trained against the pre-contrast volume; an <code>--overwrite</code> pass resurrected 729 pruned non-enhancers

A 28-assertion preflight (`python -m tier1_static.selfcheck`) now pins every one of these so they cannot regress silently.

8. What we can and cannot claim

Run-to-run variance is real and was measured. Two config-identical runs at the same seed differ:

metric	delta between identical runs
roi_pearson	0.017
roi_var_ratio	0.055
p75_corr	0.276

Training is not deterministic (cuDNN autotuning, non-deterministic 3D conv atomics), adversarial dynamics amplify it, and best-checkpoint selection turns a bumpy validation curve into a noisy discrete choice of epoch.

Rule of thumb: differences under ~ 0.02 roi_pearson are unresolved at $n = 1$.

- **Safe to claim:** GAN > flow (0.113 gap, $\sim 7\times$ noise); model » t2w baseline ($\sim 30\times$); PZ < TZ (paired within run); best > last checkpoint ($\sim 4\times$).
- **Do not claim:** anything from p75_corr at $n = 1$; the iso-vs-no-iso comparison (0.020 — inside noise; iso is justified on correctness grounds, not measured gain).

A deeper observation worth making: *reproducibility tracks identifiability*. The component the inputs determine (structure) reproduces at ± 0.017 ; the component they don't (amplitude) drifts freely, because nothing in the loss pins it down. The instability is a measurement of which part of the task is ill-posed.

Other caveats. 2D and 3D ROI metrics aren't comparable (2D keeps only gland-containing slices, so the denominators differ). FID was measured on the last checkpoint, not the best. Everything predating the audit is superseded.

9. Questions you should expect

"Is it just copying the T2w?" No — T2w-copy scores roi_pearson -0.043 against the model's $+0.482$. That baseline is in the paper.

"Why not just report SSIM/PSNR?" They're $\sim 99\%$ background here. A model can score well globally while reconstructing the gland poorly; the global-vs-ROI gap is the diagnostic.

"Is 0.48 good?" It is far above every trivial baseline and below what the task would allow if amplitude were recoverable. We've decomposed exactly which part is missing and why.

"Why is the GAN beating the diffusion/flow model?" Consistently, across four setups, at this data scale. The flow produces more realistic *texture* but localizes worse — a faithfulness/realism tradeoff, not a bug.

"Will more data fix it?" Unknown, and we're careful not to claim otherwise — an earlier "data scale doesn't help" conclusion was measured on a configuration with known bugs and has been retracted.

"Can this replace real DCE?" Not on this evidence. Per-patient enhancement amplitude isn't recoverable, and PZ — where it matters most — is the weakest region. The defensible claim is about curve *shape*, which is scale-invariant.

10. What's next

	status
Pre-contrast conditioning (--use-pregad)	running — tests whether amplitude is recoverable from the pre-contrast baseline. Decisive, since the protocol metadata is gone
PI-RADS / DCE-positivity labels	waiting on Hanxue; unlocks the clinical endpoint and a downstream readout
Multi-timepoint staging	code landed; anchors on a common time grid, goal state at 240 s (91.9% coverage; 300 s reaches only 49.7%)

	status
Tier-2 / kinetics	ODEWorld-style continuous-time model with the velocity field constrained to the Tofts equation rather than learned free-form — we know the governing ODE, which the method's own related work calls out as the limiting requirement
Multi-seed repeats	needed before any marginal comparison is publishable